

Area of Triangles

Lesson 5-3

Name: _____

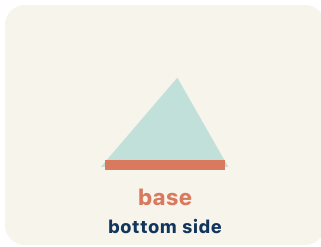
Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

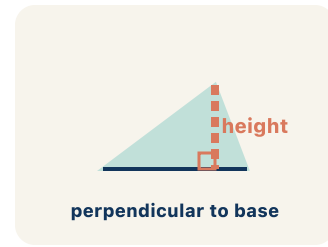
Picture first, then the word, then a plain-language meaning. Say each word out loud.



*If the bottom of a triangle is 10 cm, then $b = 10$ cm
in the formula $A = \frac{1}{2} \times b \times h$*

Base

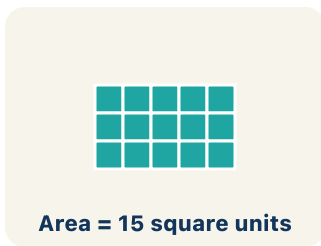
Write the definition:



*A dashed vertical line from the top point straight
down to the base at a 90° angle — like dropping a
plumb line*

Height

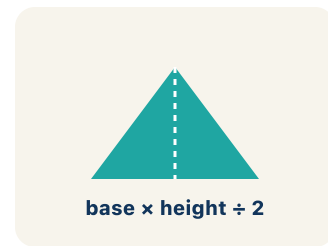
Write the definition:



*A triangle with $b = 8$ and $h = 6$ has area = $\frac{1}{2} \times 8 \times$
 $6 = 24$ sq units — exactly half the rectangle around
it*

Area

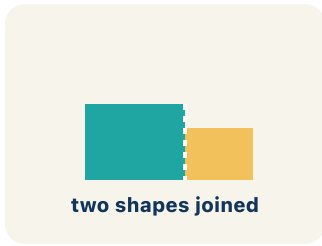
Write the definition:



*The corner of a book or a door frame — the edges
meet at exactly 90° , shown by a small square
symbol*

Perpendicular

Write the definition:



two shapes joined

A house shape = a rectangle (the walls) + a triangle (the roof); total area = rectangle area + triangle area

Composite figure

Write the definition:

$$3x + 5$$

no equal sign

$A = \frac{1}{2} \times b \times h$ means Area equals one-half times base times height; for $b = 12$ and $h = 8$, $A = 48$

Formula

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can find the area of a triangle using the formula $A = \frac{1}{2} \times \text{base} \times \text{height}$.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Base

Height

Area

Perpendicular

Composite figure

Formula

1 The side of a triangle used as the bottom for measuring area is the _____.

2 The perpendicular distance from the base to the opposite vertex is the _____.

3 The amount of flat space inside a triangle is its _____.

4 Two lines that meet at a 90° angle are _____.

5 A shape made of two or more basic shapes combined is a _____.

6 A math rule written with symbols, like $A = \frac{1}{2} \times b \times h$, is a _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: What is the area of a triangle with base 10 cm and height 6 cm?

- A. 30 sq cm
- B. 60 sq cm
- C. 16 sq cm
- D. 30 cm

Step 1 $A = \frac{1}{2} \times b \times h = \frac{1}{2} \times 10 \times 6 = 30$ square centimeters.

 **Answer:** A. 30 sq cm

 We do — together

Solve this one with your class using the same steps.

Problem: A triangle has an area of 24 sq ft and a base of 8 ft. What is the height?

- A. 6 ft
- B. 3 ft
- C. 16 ft
- D. 12 ft

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Why do we divide by 2 when finding the area of a triangle?

- A. A triangle is exactly half of a rectangle with the same base and height
- B. Triangles have 2 equal sides
- C. The base is always twice the height
- D. We always divide area formulas by 2

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. A triangular roof panel on the blueprint has a base of 7 ft and a height of 4 ft. What is its area?

- A. 14 sq ft
- B. 28 sq ft
- C. 11 sq ft
- D. 22 sq ft

Show your work:

2. On the blueprint, a triangular garden bed has a base of 11 ft and a height of 6 ft. What is its area?

- A. 33 sq ft
- B. 66 sq ft
- C. 17 sq ft
- D. 30 sq ft

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A rectangle is 12 ft by 8 ft. A diagonal line cuts it into two triangles. What is the area of each triangle? Explain how the triangle area formula relates to the rectangle area formula.

Sentence starter: The rectangle's area is ____ because _____. Each triangle's area is ____ because a diagonal splits a rectangle into ____ equal triangles, so $A = \frac{1}{2} \times \text{____} \times \text{____} = \text{_____}$.

Show your work:

Reflect — Exit Ticket

A triangle has a base of 11 inches and a height of 8 inches. What is its area?

- A. 44 sq in
- B. 88 sq in
- C. 19 sq in
- D. 44 in

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Base
2. **Notes 2:** Height
3. **Notes 3:** Area
4. **Notes 4:** Perpendicular
5. **Notes 5:** Composite figure
6. **Notes 6:** Formula
7. **We Try (notes):** A. $6 \text{ ft} - A = \frac{1}{2} \times b \times h \rightarrow 24 = \frac{1}{2} \times 8 \times h \rightarrow 24 = 4h \rightarrow h = 6 \text{ ft}.$
8. **You Try (notes):** A. A triangle is exactly half of a rectangle with the same base and height — *A triangle with base b and height h fits inside a rectangle with the same base and height. The triangle covers exactly half the rectangle, so $A = \frac{1}{2} \times b \times h.$*
9. **Try It 1:** A. $14 \text{ sq ft} - A = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 7 \times 4 = \frac{1}{2} \times 28 = 14 \text{ sq ft}.$ *Multiply base $\times$ height first (28), then take half (14).*
10. **Try It 2:** A. $33 \text{ sq ft} - A = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 11 \times 6 = \frac{1}{2} \times 66 = 33 \text{ sq ft}.$
11. **Exit Ticket:** A. $44 \text{ sq in} - A = \frac{1}{2} \times 11 \times 8 = \frac{1}{2} \times 88 = 44 \text{ square inches}.$